MATH4515 Statistical and Computational Methods in Finance
Suggested Solution of Problem Set 1

Problem 1

We let X be a random variable and let p be the probability that X is greater or equal to a fixed number $c > 0$, i.e. $P(X \geq c)$.

To estimate p, a random sample of size n is drawn from the distribution of X. Given the random sample, we estimate p by

$$\hat{p}_n = \frac{Number\ of\ X_i s\ \ with\ X_i \geq c}{n}.$$

(a) Determine if $\hat{p}_n$ is unbiased estimator.

(b) Determine if $\hat{p}_n$ is consistent estimator.

(c) Calculate the mean square error of the estimator $\hat{p}_n$.

Solution

(a) Note that $\hat{p}_n$ can be expressed as

$$\hat{p}_n = \frac{\mathbf{1}_{\{X_1 \geq c\}} + \mathbf{1}_{\{X_2 \geq c\}} + \cdots + \mathbf{1}_{\{X_n \geq c\}}}{n}.$$

Here, $\mathbf{1}_{\{X_i \geq c\}} = \begin{cases} 1 & if\ \ X_i \geq c \\ 0 & if\ \ X_i < c \end{cases}$.

Recall that $p = P(X_i > c)$. It follows that

$$\mathbb{E}\left[\mathbf{1}_{\{X_i > c\}}\right] = 1 \times p + 0 \times (1-p) = p$$

and

$$\mathbb{E}[\hat{p}_n] = \frac{1}{n}\sum_{i=1}^{n} \underbrace{\mathbb{E}\left[\mathbf{1}_{\{X_i > c\}}\right]}_{=p} = \frac{np}{n} = p.$$

So $\hat{p}_n$ is unbiased.

(b) When $n \to \infty$, it follows from Central Limit Theorem that $\hat{p}_n = \frac{\mathbf{1}_{\{X_1 \geq c\}} + \mathbf{1}_{\{X_2 \geq c\}} + \cdots + \mathbf{1}_{\{X_n \geq c\}}}{n}$ is approximately normal with mean $\mu = \mathbb{E}\left[\mathbf{1}_{\{X_i > c\}}\right] = p$ and variance $\frac{\sigma^2}{n} = \frac{p(1-p)}{n}$, where $\sigma^2 = Var\left(\mathbf{1}_{\{X_i > c\}}\right) = \underbrace{\mathbb{E}\left[\left(\mathbf{1}_{\{X_i > c\}}\right)^2\right]}_{=1^2 \times p + 0^2 \times (1-p)} - \left(\mathbb{E}\left[\mathbf{1}_{\{X_i > c\}}\right]\right)^2 = p(1-p)$.

Then we deduce that for any $\varepsilon > 0$

$$\lim_{n\to\infty} P(|\hat{p}_n - p| > \varepsilon) = \lim_{n\to\infty} [P(\hat{p}_n < p - \varepsilon) + P(\hat{p}_n > p + \varepsilon)]$$

$$= \lim_{n\to\infty} \left[P\left(Z < -\frac{\varepsilon}{\sqrt{\frac{p(1-p)}{n}}} \right) + P\left(Z > \frac{\varepsilon}{\sqrt{\frac{p(1-p)}{n}}} \right) \right]$$

$$= \lim_{n\to\infty} \left(\int_{-\infty}^{-\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz + \int_{\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \right) = 0 + 0 = 0$$

So $\hat{p}_n$ is consistent.

(c) Recall that the mean square error of $\hat{p}_n$ is

$$MSE(\hat{p}_n) = Var(\hat{p}_n) + \big(Bia(\hat{p}_n)\big)^2.$$

Since $\hat{p}_n$ is unbiased, so $Bia(\hat{p}_n) = 0$. So it follows from the result in (b) that

$$MSE(\hat{p}_n) = Var(\hat{p}_n) = \frac{p(1-p)}{n}.$$

Problem 2

We let X be a random variable with the following probability density function:

$$f_X(x) = \frac{2x}{\theta^2}, \qquad 0 \le x \le \theta.$$

To determine θ, a random sample of size n is taken from the distribution of X. We consider the point estimator $\hat{\theta}_n = \max(X_1, X_2, \dots, X_n)$.

(a) Determine if it is unbiased estimator. If not, derive another estimator which is unbiased.

(b) Determine if $\hat{\theta}_n$ is consistent estimator.

(c) Calculate the mean square error of the estimator $\hat{\theta}_n$.

☺Solution

To facilitate the analysis, we first compute the probability density function of $\max(X_1, X_2, \dots, X_n)$.

For any $0 \le x \le \theta$, we have

$$F_{\hat{\theta}_n}(x) = P(\max(X_1, X_2, \dots, X_n) \le x) = P(X_i \le x \ \ for\ all\ i = 1,2, \dots, n)$$

$$= \prod_{i=1}^{n} P(X_i \le x) = \prod_{i=1}^{n} \int_0^x \frac{2y}{\theta^2} dy = \prod_{i=1}^{n} \left(\frac{x}{\theta}\right)^2 = \frac{x^{2n}}{\theta^{2n}}.$$

Thus, the probability density function of $\hat{\theta}_n$ is found to be

$$f_{\hat{\theta}_n}(x) = F'_{\hat{\theta}_n}(x) = \frac{2nx^{2n-1}}{\theta^{2n}}, \qquad 0 \le x \le \theta.$$

(a) By direct calculation, we get

$$\int_0^\theta x\left(\frac{2nx^{2n-1}}{\theta^{2n}}\right) dx = \int_0^\theta \frac{2nx^{2n}}{\theta^{2n}} dz = \left[\frac{2nx^{2n+1}}{\theta^{2n}(2n+1)}\right]_0^\theta = \frac{2n}{2n+1}\theta \ne \theta.$$

So the estimator is unbiased.

On the other hand, one can consider the estimator

$$\hat{\theta}_n^{new} = \frac{2n+1}{2n}\max(X_1, X_2, \dots, X_n).$$

One can verify that $\mathbb{E}\big[\hat{\theta}_n^{new}\big] = \theta$ and this new estimator is unbiased.

(b) For any $\varepsilon > 0$, we have

$$\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| > \varepsilon\right) = \lim_{n\to\infty}\left[P\left(\hat{\theta}_n < \theta - \varepsilon\right) + \underbrace{P\left(\hat{\theta}_n > \theta + \varepsilon\right)}_{=0 \ as\ \hat{\theta}_n \le \theta}\right]$$

$$= \lim_{n\to\infty}\int_0^{\theta-\varepsilon} \frac{2nx^{2n-1}}{\theta^{2n}}\,dx = \lim_{n\to\infty}\left(\frac{\theta-\varepsilon}{\theta}\right)^{2n} = 0.$$

So $\hat{\theta}_n$ is consistent.

(c) Note that

$$Bia\left(\hat{\theta}_n\right) = \mathbb{E}\left[\hat{\theta}_n\right] - \theta = \frac{2n}{2n+1}\theta - \theta = -\frac{\theta}{2n+1}.$$

On the other hand,

$$\mathbb{E}\left[\left(\hat{\theta}_n\right)^2\right] = \int_0^{\theta} x^2\left(\frac{2nx^{2n-1}}{\theta^{2n}}\right) = \int_0^{\theta}\frac{2nx^{2n+1}}{\theta^{2n}}\,dz = \left[\frac{2nx^{2n+2}}{\theta^{2n}(2n+2)}\right]_0^{\theta}$$

$$= \frac{2n}{2n+2}\theta^2$$

Hence, we deduce that

$$Var\left(\hat{\theta}_n\right) = \underbrace{\frac{2n}{2n+2}\theta^2 - \left(\frac{2n}{2n+1}\theta\right)^2}_{\mathbb{E}\left[\hat{\theta}_n^2\right] - \left(\mathbb{E}\left[\hat{\theta}_n\right]\right)^2} = \frac{2n\theta^2}{(2n+2)(2n+1)^2}.$$

The mean square error is found to be

$$MSE\left(\hat{\theta}_n\right) = \underbrace{\frac{2n\theta^2}{(2n+2)(2n+1)^2} + \left(-\frac{\theta}{2n+1}\right)^2}_{Var\left(\hat{\theta}_n\right) + \left(Bia\left(\hat{\theta}_n\right)\right)^2} = \frac{2\theta^2}{(2n+2)(2n+1)}.$$

Problem 3

(a) We let N be a random variable which has Binomial distribution with parameters $n$and $p \in [0,1]$. That is, the probability mass function of N is given by

$$P(N = k) = C_k^n p^k (1-p)^{n-k}, \qquad for\ \ k = 0,1,2,\dots,n$$

Here, n is a fixed positive integer (assumed to be known).

To determine p, a random sample of size m (denoted by $\{N_1, N_2, \dots, N_m\}$) is taken.

(i) Determine the maximum likelihood estimate of p. Express your answer in terms of $N_1, N_2, \dots, N_m$.

(ii) Hence, estimate the mean square error of the MLE obtained in (a)(i).

(b) We let X be a continuous random variable which has the following probability density function:

$$f_X(x) = \frac{\theta}{x^2}, \qquad x \geq \theta,$$

where $\theta > 0$ is constant. To determine θ, a random sample of size n (denoted by $\{X_1, X_2 \ldots, X_n\}$) is taken.

(i) Determine the maximum likelihood estimate of θ. (😊 Hint: Express your answer in terms of $X_1, X_2, \ldots, X_n$. Be careful that θ also affects the range of X.)

(ii) Hence, estimate the mean square error of the MLE obtained in (b)(i) using suitable method.

😊Solution

(a) Firstly, the likelihood function is given by

$$L(p) = L(p|N_1, \ldots, N_m) = \prod_{i=1}^{m} P(N_i = n_i) = \prod_{i=1}^{m} C_{n_i}^{10} p^{n_i}(1-p)^{n-n_i}$$
$$= \left(\prod_{i=1}^{m} C_{n_1}^{n}\right) p^{\sum_{i=1}^{m} n_i}(1-p)^{nm-\sum_{i-=1}^{m} n_i}.$$

In order to find p that maximize the likelihood function, we consider the loglikelihood function:

$$l(p) = \ln L(p) = \ln\left(\prod_{i=1}^{m} C_{n_1}^{n}\right) + \left(\sum_{i=1}^{m} n_i\right)\ln p + \left(nm - \sum_{i-=1}^{m} n_i\right)\ln(1-p)$$

To find the maximizer, we consider the first order condition:

$$\frac{d}{dp} l(p) = 0 \Rightarrow \frac{1}{p}\left(\sum_{i=1}^{m} n_i\right) - \frac{1}{1-p}\left(nm - \sum_{i=1}^{m} n_i\right) = 0 \Rightarrow \hat{p} = \frac{\sum_{i=1}^{m} n_i}{nm}.$$

One can show that

$$\frac{d^2}{dp^2} l(p) = -\frac{1}{p^2}\left(\sum_{i=1}^{m} n_i\right) - \frac{1}{(1-p)^2}\underbrace{\left(nm - \sum_{i=1}^{m} n_i\right)}_{\geq 0} < 0.$$

So $l(p)$ is maximized at $p = \hat{p}$ and $\hat{p} = \frac{\sum_{i=1}^{m} n_i}{nm}$ is the desired MLE.

(b) Note that θ represents the *lowest possible value* of the random variable X, then the corresponding likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f_X(X_i) = \begin{cases} \prod_{i=1}^{n} \frac{\theta}{X_i^2} = \frac{\theta^n}{\prod_{i=1}^{n} X_i^2} & if \ \ \theta \leq \min(X_1, X_2, \ldots, X_n) \\ 0 & if \ \ \theta > \min(X_1, X_2, \ldots, X_n) \end{cases}.$$

Since $\frac{\theta^n}{\prod_{i=1}^{n} X_i^2}$ is increasing in θ when $\theta \leq \min(X_1, X_2, \ldots, X_n)$, it follows that $L(\theta)$ is maximized at $\theta = \hat{\theta} = \min(X_1, X_2, \ldots, X_n)$ and $\hat{\theta}$ is the desired MLE.

Problem 4

We let X be a random variable which has the following probability density function

$$f_X(x) = \begin{cases} \dfrac{(\alpha+1)x^\alpha}{\theta^{\alpha+1}} & if\ \ 0 \le x \le \theta \\ 0 & if\ otherwise \end{cases} .$$

Here, α and θ are positive parameters.

To estimate the parameter, a random sample of size n $\{X_1, X_2, \dots, X_n\}$ is drawn from the distribution.

(a) Suppose that θ is known, estimate α using maximum likelihood method.

(b) Suppose that α is known, estimate θ using maximum likelihood method.

(c) (Harder) Suppose that both θ and α are unknown, determine the maximum likelihood method.

(☺Hint: You can first focus on predicting the MLE of one of the parameters)

😊Solution

(a) Since α does not affect the range of the distribution, one can determine the MLE using the standard method.

Note that the loglikelihood function is given by

$$l(\alpha) = \ln L(\alpha) = \sum_{i=1}^{n} \underbrace{\ln \frac{(\alpha+1)X_i^\alpha}{\theta^{\alpha+1}}}_{\ln f_X(X_i)}$$

$$= n\ln(\alpha+1) - \alpha\sum_{i=1}^{n} \ln X_i - n(\alpha+1)\ln\theta.$$

Next, we consider the first order condition:

$$\frac{\partial l}{\partial \alpha} = 0 \Rightarrow \frac{n}{\alpha+1} - \sum_{i=1}^{n} \ln X_i - n\ln\theta = 0 \Rightarrow \alpha = \frac{n}{\sum_{i=1}^{n} \ln X_i + n\ln\theta} - 1.$$

One can check that $\frac{\partial^2 l}{\partial \alpha^2} = -\frac{n}{(\alpha+1)^2} < 0$. So the MLE of α is

$$\hat{\alpha}_n = \frac{n}{\sum_{i=1}^{n} \ln X_i + n\ln\theta} - 1.$$

(b) One has to be careful that θ affects the range of the random variable. Given $\{X_1, X_2, \dots, X_n\}$, the likelihood function is given by

$$L(\theta) = \prod_{i=1}^{n} f_X(X_i|\theta)$$

$$= \begin{cases} \prod_{i=1}^{n} \dfrac{(\alpha+1)X_i^\alpha}{\theta^{\alpha+1}} = \dfrac{(\alpha+1)^n(\prod_{i=1}^{n} X_i)^\alpha}{\theta^{n(\alpha+1)}}, & if\ \theta \ge \max(X_1, \dots, X_n) \\ 0 & if\ \theta < \max(X_1, \dots, X_n) \end{cases}$$

Note that $L(\theta) = \frac{(\alpha+1)^n(\prod_{i=1}^{n} X_i)^\alpha}{\theta^{n(\alpha+1)}} > 0$ is decreasing with respect to θ for $\theta \ge \max(X_1, \dots, X_n)$. As $L(\theta) = 0$ for $\theta < \max(X_1, \dots, X_n)$, so $L(\theta)$ is maximized when $\theta = \max(X_1, X_2, \dots, X_n)$.

Therefore, the MLE of θ is

$$\hat{\theta}_n = \max(X_1, X_2, \dots, X_n).$$

(c) The likelihood function is given by

$$L(\alpha, \theta) = \begin{cases} \prod_{i=1}^{n} \frac{(\alpha+1)X_i^{\alpha}}{\theta^{\alpha+1}} = \frac{(\alpha+1)^n (\prod_{i=1}^{n} X_i)^{\alpha}}{\theta^{n(\alpha+1)}}, & if\ \theta \geq \max(X_1, \dots, X_n) \\ 0 & if\ \theta < \max(X_1, \dots, X_n) \end{cases}.$$

For any $\alpha > 0$, we observe from the result of (b) that L is maximized at $\hat{\theta}_n = \max(X_1, X_2, \dots, X_n)$.

Then the corresponding likelihood function can be expressed as

$$L(\alpha) = \frac{(\alpha+1)^n (\prod_{i=1}^{n} X_i)^{\alpha}}{\hat{\theta}_n{}^{n(\alpha+1)}}$$

Since $\hat{\theta}_n$ is independent of α, thus the MLE of α is same as that in (a), i.e.

$$\hat{\alpha}_n = \frac{n}{\sum_{i=1}^{n} \ln X_i + n \ln \hat{\theta}_n} - 1.$$

Problem 5

In this problem, we consider the daily return of a stock. The excel file "StockA.xlsx" the historical daily return of the stock from 20th Sept, 2010 to 17th June, 2019. Complete the following task:

- **(a)** We let μ be the expected daily return of the stock. Find 98% confidence interval of μ.
- **(b)** We let p be the probability that the stock price drops by more than 2% over a trading day. Determine if $p > 15\%$ using hypothesis test. Take the significant level to be $\alpha = 1\%$.
- **(c)** We let σ be the standard deviation of the stock daily return. Determine if $\sigma > 3\%$. Take the significant level to be $\alpha = 5\%$.

Provide full justification to your answer.

(*To facilitate the analysis, we assume that the daily return of the stock follows normal distribution)

☺Solution

(a) We let $\bar{r}$ and s^2 be the sample mean and sample variance of the daily return of the stock. Using the given data in excel, we can deduce that

$$\bar{r} = 0.001588408, \qquad s = 0.03212659.$$

Assuming that the daily return follows normal distribution, it follows from the result in Example 6 that $T = \frac{\bar{r}-\mu}{s/\sqrt{n}}$ follows student-t distribution with degree of freedom $n - 1 = 2198$.

Next, we find the critical values t_L and t_U such that

$$P\left(t_L \le T = \frac{\bar{r}-\mu}{\frac{s}{\sqrt{n}}} \le t_U\right) = 0.98$$

By taking $P(T \le t_L) = 0.01$ and $P(T \le t_U) = 0.99$, we get

$$t_L = -2.328046, \qquad t_U = 2.328046.$$

Then we deduce that

$$P\left(-2.328046 \le T = \frac{\bar{r}-\mu}{\frac{s}{\sqrt{2199}}} \le 2.328046\right) = 0.98$$

$$\Rightarrow P\left(\bar{r} - 2.328046\frac{s}{\sqrt{2199}} \le \mu \le \bar{r} + 2.328046\frac{s}{\sqrt{2199}}\right) = 0.98$$

So the 98% confidence interval is found to be

$$\left[\bar{r} - 2.328046\frac{s}{\sqrt{2199}},\ \ \bar{r} + 2.328046\frac{s}{\sqrt{2199}}\right]$$
$$= [-6.529199 \times 10^{-6},\ \ 0.003183345]$$

(b) We let p be the probability that the stock price drops by more than 2% in a trading day. Using the daily return of Tesla stock, p can be estimated as

$$\hat{p} = \frac{\mathbf{1}_{\{r_1<-2\%\}} + \mathbf{1}_{\{r_2<-2\%\}} + \cdots + 1_{\{r_n<-2\%\}}}{n} \approx 0.1914507.$$

To determine whether $p > 15\%$, we adopt the hypothesis testing (with $\alpha = 1\%$) as follows;

- Step 1: The hypotheses are set to be
$$\begin{cases} H_0: p \le 15\% \\ H_1: p > 15\% \end{cases}$$
Since one-tail test is adopted, one should accept H_0 when $\hat{p} \le p^*$ and accept H_1 when $\hat{p} > p^*$.
- Step 2: By the central limit theorem, $\hat{p}$ approximates follows normal distribution with mean p and variance $\frac{p(1-p)}{n}$ since the sample size $n = 2199$ is sufficiently large. Then
$$Z = \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1).$$
- Step 3: Given the significant level $\alpha = 1\%$, we choose Z^* such that
$$P\left(Z = \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \ge Z^*\right) = 0.01 \Rightarrow Z^* = 2.326348.$$
- Step 4: Assuming H_0 is true and $p = 0.15$, then the exact value of the test statistic is
$$Z = \frac{0.1914507 - 0.15}{\sqrt{\frac{0.15(1-0.15)}{2199}}} \approx 5.44364 > 2.326348.$$

Therefore, we shall accept H_1 and conclude that $p > 15\%$.

(c) According to the result in (a), the sample standard deviation of the daily return is

$$s = 0.03212659.$$

To determine whether $\sigma > 3\%$. , we adopt the hypothesis testing (with $\alpha = 1\%$) as follows;

- Step 1: The hypotheses are set to be
$$\begin{cases} H_0: \sigma = 3\% \\ H_1: \sigma > 3\% \end{cases} \left(or \begin{cases} H_0: \sigma \leq 3\% \\ H_1: \sigma > 3\% \end{cases} \right)$$
Since one-tail test is adopted, one should accept H_0 when $s \leq s^*$ and accept H_1 when $s > s^*$.
- Step 2: Assuming the stock return is normally distributed, it follows that the test statistics $\frac{(n-1)s^2}{\sigma^2}$ has chi-squared distribution with degree of freedom $n - 1 = 2198$.
- Step 3: Given the significant level $\alpha = 5\%$, we choose Z^* such that
$$P\left(\frac{(n-1)s^2}{\sigma^2} \geq Z^*\right) = 0.05 \Rightarrow Z^* = 2308.183.$$
- Step 4: Assuming H_0 is true which $\sigma = 0.03$, then the exact value of the test statistic is
$$\frac{(n-1)s^2}{\sigma^2} = \frac{2198(0.03212659)^2}{(0.03)^2} \approx 2520.661 > 2308.183.$$
Therefore, we shall accept H_1 and conclude that $\sigma > 3\%$.

Problem 6

An analyst claims that a random variable X has the following probability density function:

$$f_X(x) = \frac{2}{(1+x)^3}, \qquad x > 0.$$

To test the claim, a random sample of size 10 is obtain. The sample data is presented below:

$$\{0.1,\ 0.2,\ 0.4,\ 0.5,\ 0.7,\ 0.75,\ 0.9,\ 1,\ 1.3,\ 1.6\}.$$

Suppose that the analyst would like to test the hypothesis using Kolmogorov-Smirnov test, calculate the Kolmogorov-Smirnov test statistics.

😊Solution

To facilitate the calculation, we first derive the formula of the cumulative distribution function. That is,

$$F_x(x) = P(X \leq x) = \int_0^x \frac{2}{(1+y)^3} dy = \left[-\frac{1}{(1+y)^2}\right]_0^x = 1 - \frac{1}{(1+x)^2}.$$

One can calculate the KS-test through the following formula:

$$D = \sup_{i=1,2,\ldots n+1} |\max(|F_n(X_{i-1}) - F_X(X_{i-1})|, |F_n(X_{i-1}) - F_X(X_i)|)|$$

where $\{X_1, X_2, \ldots, X_n\}$ denotes the random sample. (*Note: Here, we take $X_0 = -\infty$ and $X_{n+1} = +\infty$)

The sample R-code is presented as follows:

```
x=c(0.1, 0.2, 0.4, 0.5, 0.75, 0.9, 1, 1.3, 1.6) #Input the data

n=length(x) #Calculate the sample size of the data

#Write a function to calculuate the cdf of exponential distribution
cdfnorm<-function(x){
 cdf=1-1/((1+x)^2)
 return(cdf)
}

Dint=abs(cdfnorm(x[1]))

Dtest=rep(0,n) #Set an array to store the values of |Fn-FX| for calculuaing

for (i in 1:n){
 Fn=sum(ifelse(x<=x[i],1,0))/n; #Calculate the empirical distribution
 D1test=abs(Fn-cdfnorm(x[i])) #For calculating |Fn(X_i)-FX(X_i)|
 if (i<n){
  D2test=abs(Fn-cdfnorm(x[i+1])) #For calculating |Fn(X_i)-FX(X_(i+1))|
 } else if (i==n){
  D2test=0
 }
 Dtest[i]=max(D1test,D2test)
}

D=max(Dint,max(Dtest))
print(D)
```

The D-value is found to be **0.2675737**.

Problem 7

We adopt the setup as in Problem 6, calculate the test statistic of Anderson-Darling test.

😊Solution

One can calculate the test statistics using the formula below:

$$D_n^* = -\sum_{k=1}^{n} \frac{2k-1}{n} \left[\ln F_X(X_k) + \ln\left(1 - F_X(X_{n+1-k})\right)\right] - n.$$

The sample R-code is given as follows:

```
x=c(0.1, 0.2, 0.4, 0.5, 0.75, 0.9, 1, 1.3, 1.6) #Input the data

n=length(x) #Calculate the sample size of the data

#Write a function to calculuate the cdf of exponential distribution
cdfX<-function(x){
  cdf=1-1/((1+x)^2)
  return(cdf)
}

Dterm=rep(0,n) #Set an array to store the terms in the summations

for (k in 1:n){
  Dterm[k]=((2*k-1)/n)*(log(cdfX(x[k]))+log(1-cdfX(x[n+1-k]))) #Calculuate the
terms in the summation
}

D=-sum(Dterm)-n #Calculate the test statistic
print(D)
```

The value of the test statistics is found to be **0.888801**.

Problem 8

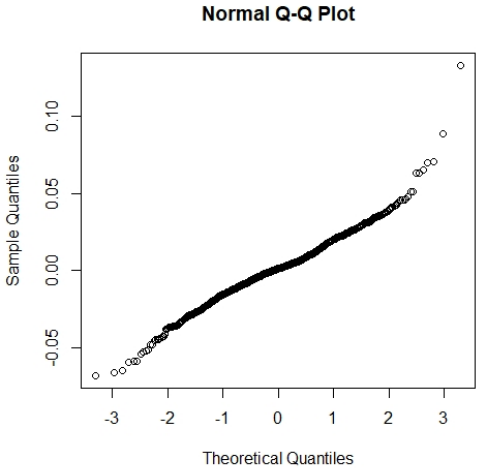
We consider three stocks traded in certain market (StockA, StockB, StockC).
In this problem, we would like to examine whether the daily logreturn of three stock follows normal distribution and t-distribution by analyzing the historical returns of these stocks see the excel file "StockABC_Logreturn" which is available in canvas).
Complete the following task using *RStudio*:

- **(a)** Firstly, we investigate whether the daily return of these stock follows normal distribution.
  - **(i)** For each of the stock, plot the qq-plot against normal distribution. Give the conclusion based on the observation.
  - **(ii)** Perform the KS-test and Anderson-Darling test for each stock and give the conclusion. Here, we take the significant level to be $\alpha = 0.01$.
- **(b)** Secondly, we investigate whether the daily return of these stock follows general t-distribution.
  - **(i)** For each of the stock, plot the qq-plot against t-distribution and give the conclusion.
  - **(ii)** Perform the KS-test and Anderson-Darling test for each stock and give the conclusion. Here, we take the significant level to be $\alpha = 0.05$.

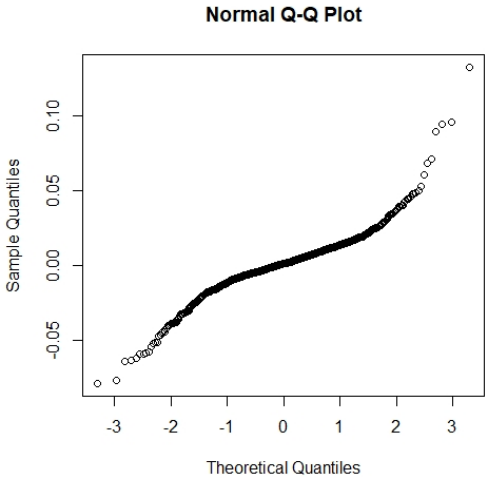
☺Solution

(a) (i) The following is the QQplots of those three stocks against the standard normal distribution (which can be done using qqnorm(x), where x denotes the historical daily return of the stock)

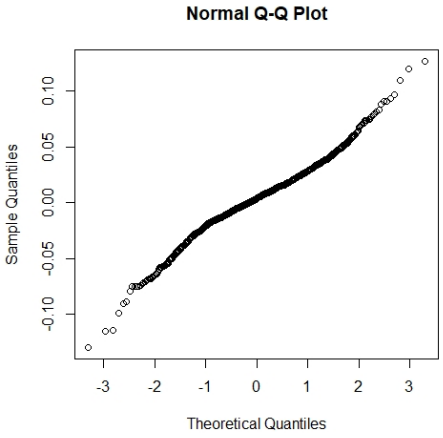
StockA



StockB



StockC



Comment: Since the curves of these three stocks are not straight line, it follows that the stock returns may not be normally distributed.

(ii) Next we shall consider the normality using KS-test and AD-test. The sample code and sample output are given as follows (*Note: Here, I take "StockA" as an example)

```
library(MASS)

StockABC_Logreturn<-na.omit(StockABC_Logreturn)
x<-StockABC_Logreturn$StockA
x<-x+rnorm(length(x),mean=0,sd=1)*0.00000000000000000001 #Avo
iding tie in data

MLEStockA<-fitdistr(x,"normal") #Compute the MLEs of the para
meters (normal distribution)

ks.test(x,"pnorm",mean=MLEStockA$estimate["mean"],sd=MLEStock
A$estimate["sd"]) #Performing KS-test

        Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.050137, p-value = 0.01285
alternative hypothesis: two-sided


library(goftest)
ad.test(x,"pnorm",mean=MLEStockA$estimate["mean"],sd=MLEStock
A$estimate["sd"]) #Performing AD-test

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Normal distribution
        with parameters mean = 0.00120856982383527, sd = 0.01
95840457945454
        Parameters assumed to be fixed

data:  x
An = 3.9488, p-value = 0.009241
```

Based on the value of p-value, we shall reject H_0 (the stock return is normally distributed) and accept H_1 (the stock return is not normally distributed) only when the p-value is lower than the significant level $\alpha = 0.01$.

The following tables show the p-values of those three stocks under KS-test and AD-test

	KS-test	AD-test
StockA	0.01285	0.009241
StockB	3.241×10^{-8}	5.976×10^{-7}
StockC	0.001567	0.001058

Comments

- For KS.test, note that the p-values of AMZN and SHOP are both less than $\alpha = 0.01$, it follows that the stock returns of AMZN and SHOP are not normally distributed. On the other hand, the p-value of BABA is slightly greater than $\alpha = 0.01$. So there is no sufficient evidence that the stock return of BABA is normally distributed.

- For AD-test, note that p-values of all these 3 stocks are less than $\alpha = 0.01$, it follows that the stock returns of these three stocks are not normally distributed.

(b) To facilitate the analysis, we first determine the MLEs of the parameters μ (mean parameter), λ (scale parameter) and ν (degree of freedom) of each of these three stocks when they are fit into general t-distribution. The R-code for determining MLE is given as follows (Here, we take Stock A as an example):

```
library("Rfast")
library("ramify")
library("LambertW")

x<-StockABC_Logreturn$StockB #Store the target data
x<-na.omit(x)            #remove "NA" in the dataset
x<-as.matrix(x)          #ensure that the data is in matrix f
df=seq(2,8,0.001)        #set the range of degree of freedom
n=length(df)
loglik=rep(0,n)          #Used to store the value of loglikel
nction for different level of df
for(i in 1:n){
  fitres=mvt.mle(x,v=df[i])
  loglik[i]=fitres$loglik
}          #For each df, calculate the MLE and loglikelihoo
n
loglik<-as.matrix(loglik)
dfmax=df[argmax(loglik,rows=FALSE)] #Find the maximizier(opt
fitMLE=mvt.mle(x,v=dfmax) #Estimate the MLE of the parameter

m<-fitMLE$location            #m=mu (location)
s<-sqrt(fitMLE$scatter[1,1])  #s=lambda (scale parameter)
print(m)
print(s)
print(dfmax)
> print(m)
[1] 0.001132244 (Value of μ)
> print(s)
[1] 0.01582977  (Value of λ)
> print(dfmax)
[1] 5.729       (value of ν)
```

The following table summarizes the MLEs of the parameters of each of these three stocks

	MLE		
	μ	λ	ν
StockA	0.001132244	0.01582977	5.729
StockB	0.001574724	0.01036996	2.717
StockC	0.003734021	0.02244406	4.367

(i) Recall that if X follows general T-distirubiton, then X can be expressed as

$$X = \mu + \lambda T_{\nu},$$

where T_{ν} follows student-t distribution with degree of freedom ν.

Similar to the case for normal distribution, it follows that the distribution of X will form a straight line against student-t distribution with degree of freedom ν in the qq-plot.

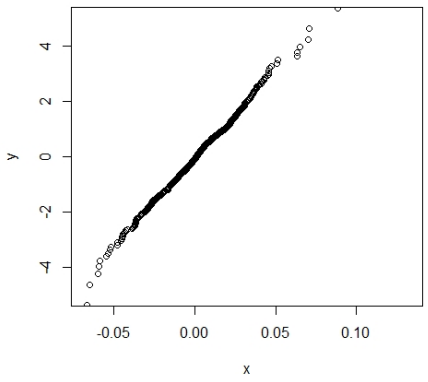
To generate the qq-plot against student-t distribution, we adopt the following R-code (*Note: I duplicate the student-t distribution by drawing a large sample from student-t distribution with given value of degree of freedom.)

```
> x<-na.omit(StockABC_Logreturn$StockA)
> y<-rt(10000000,dfmax)
> qqplot(x,y,ylim=c(-5,5))
```

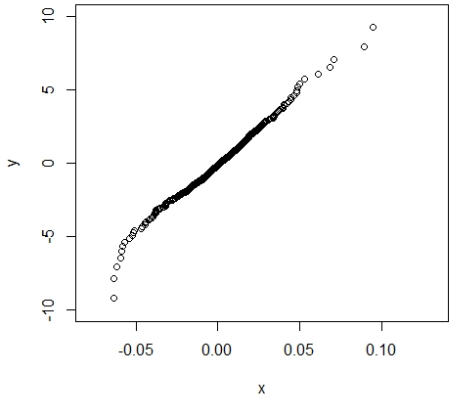
(*Note: I add the command “ylim” to control the range of y-axis of the qqplot for a clearer plot.)

The following shows the qq-plot of the stick return of these three stocks again student-t distribution:

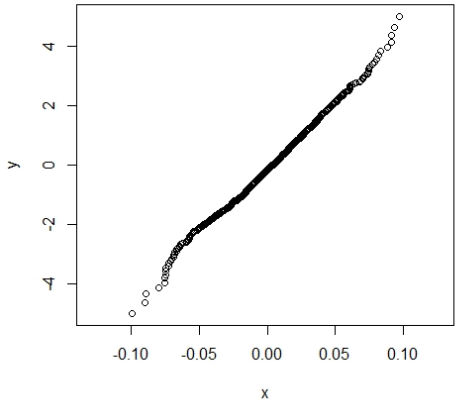
1. BABA ($\nu = 5.729$)



2. AMZN ($\nu = 2.717$)



3. SHOP ($\nu = 4.367$)



Comment

It appears that all curves roughly form straight lines in the qqplot. So it appears that these three stock returns may follows t-distirbution.

(ii) Next, we shall do the verification using KS-test and AD-test. The following are the R-code of these two tests (Here, we take Stock A as example).

```
> x<-na.omit(StockABC_Logreturn$StockA)
> m<-0.001132244
> s<-0.01582977
> T=(x-m)/s
> ks.test(T,"pt",df=dfmax) #do the testing using ks.test

        One-sample Kolmogorov-Smirnov test student-t with d
f=5.73, location=0,
        scale=0.02

data:  x
D = 0.03185, p-value = 0.2603
alternative hypothesis: two-sided
```

```
> ad.test(T,"pt",dfmax)

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Student's t distribution
        Parameters assumed to be fixed

data:  y
An = 0.84592, p-value = 0.449
```

The following tables show the p-values of those three stocks under KS-test and AD-test

	KS-test	AD-test
StockA	0.2603	0.449
StockB	0.9327	0.8163
StockC	0.9842	0.8495

We observe that all p-values are much greater than the significant level $\alpha = 0.01$ in both tests. We conclude that there is no sufficient evidence that the stock returns are not t-distributed.